

CO 2-ASSESSMENT TOOL-1

Question 1

A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyze the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system. (20 Marks)

Kali Linux Commands

`ip a`

`ping 192.168.1.10`

`nmap 192.168.1.10`

`nmap -F 192.168.1.10`

`nmap -p- 192.168.1.10`

`nmap -sV 192.168.1.10`

`nmap -O 192.168.1.10`

`nmap -A 192.168.1.10`

`nmap -A 192.168.1.10 -oN nmap_report.txt`

`cat nmap_report.txt`

Command Explanation

- `ip a` – Displays the network interfaces and IP address of the Kali Linux machine.
- `ping 192.168.1.10` – Checks whether the target system is reachable.
- `nmap 192.168.1.10` – Performs a basic scan to identify open ports.
- `nmap -F 192.168.1.10` – Scans the most common ports.

- `nmap -p- 192.168.1.10` – Scans all TCP ports (1–65535).
- `nmap -sV 192.168.1.10` – Detects the versions of services running on open ports.
- `nmap -O 192.168.1.10` – Detects the operating system.
- `nmap -A 192.168.1.10` – Performs an aggressive scan (OS detection, version detection, scripts, and traceroute).
- `nmap -A 192.168.1.10 -oN nmap_report.txt` – Saves the scan report in a text file.
- `cat nmap_report.txt` – Displays the saved scan report.

Expected Output

Host is up.

PORT	STATE	SERVICE	VERSION
22/tcp	open	ssh	OpenSSH 8.2
80/tcp	open	http	Apache 2.4
3306/tcp	open	mysql	MySQL 8.0

OS: Linux

Question 2

A phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. using the GoPhish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks. (20 Marks)

Kali Linux Commands

`cd gophish`

`ls`

`chmod +x gophish`

`./gophish`

`ps -ef | grep gophish`

`cat config.json`

`firefox https://localhost:3333`

Command Explanation

- `cd gophish` – Opens the GoPhish installation directory.
- `ls` – Displays the files available in the directory.
- `chmod +x gophish` – Gives execute permission to the GoPhish application.
- `./gophish` – Starts the GoPhish server.
- `ps -ef | grep gophish` – Checks whether the GoPhish process is running.
- `cat config.json` – Displays the GoPhish configuration file.
- `firefox https://localhost:3333` – Opens the GoPhish web dashboard.

Expected Output

Starting admin server at https://127.0.0.1:3333

Starting phishing server at http://0.0.0.0:80

Campaign Results:

Emails Sent : 100

Emails Opened : 75

Link Clicked : 40

Credentials Submitted : 20

Attachments Downloaded : 10

Question 3

A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyze the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure. (10 Marks)

Kali Linux Commands

dig -v

dig example.com

dig NS example.com

dig MX example.com

dig A example.com

dig AAAA example.com

dig TXT example.com

dig -x 192.168.1.10

dig AXFR example.com @ns1.example.com

dig AXFR example.com @ns1.example.com > dns_report.txt

cat dns_report.txt

Command Explanation

- dig -v – Displays the installed Dig version.
- dig example.com – Performs a basic DNS lookup.

- `dig NS example.com` – Displays the Name Server (NS) records.
- `dig MX example.com` – Displays the Mail Exchange (MX) records.
- `dig A example.com` – Displays the IPv4 address of the domain.
- `dig AAAA example.com` – Displays the IPv6 address of the domain.
- `dig TXT example.com` – Displays the TXT records.
- `dig -x 192.168.1.10` – Performs a reverse DNS lookup.
- `dig AXFR example.com @ns1.example.com` – Attempts a DNS Zone Transfer (AXFR).
- `dig AXFR example.com @ns1.example.com > dns_report.txt` – Saves the DNS Zone Transfer report.
- `cat dns_report.txt` – Displays the saved report.

Expected Output

`example.com.`

`www.example.com`

`mail.example.com`

`vpn.example.com`

`ftp.example.com`

`admin.example.com`

`:: Zone transfer successful.`